CURRICULUM VITAE · 2026

PROFESSIONAL EDITION

HOLLIE BUCKETS FIGUEROA

SENIOR INTERACTION DESIGNER · SYSTEMS PROTOTYPER · MODEL BEHAVIOURIST

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Portfolio & Demos Available on Request

01 EXECUTIVE PROFILE

Interaction designer, systems prototyper, and model behaviourist with over seven years at **Apple** directing advanced prototyping and human-centered design programs. Specializes in **multi-modal interactions**, **sensor architectures**, **audio-first spatial wayfinding**, and the **behavioural evaluation of generative AI foundation models**. Bridges interaction design, core software engineering, human factors research, and product incubation by translating research hypotheses into functional software prototypes, rigorous evaluation frameworks, and granted utility patents. Grounded in interactive sciences, systems management, game development, and biomimetic human-computer interaction.

02 AREAS OF EXPERTISE

INTERACTION & PROTOTYPING

Multi-Modal Interaction Design, Audio-First UX & Earcons, Spatial Audio, Sensor Integration (IMU/Acoustic/Motion), Wizard of Oz Studies, Rapid Functional Prototyping.

MODEL BEHAVIOUR & EVALUATION

Ethological Prompt Frameworks, Response Taxonomy, Demographic Drift Analysis, Adapter Sensitivity, Automated Benchmarking, LLM-as-Judge Pipelines.

INCUBATION GOVERNANCE

Stage-Gated Incubation, Automated Review Systems, Sensor Feasibility Modeling, Decision Logging, Cross-Functional Technical Roadmaps.

TECHNICAL STACK & TOOLING

Swift 6, SwiftUI, watchOS, iOS, Vision, CoreImage, CoreMotion, SoundAnalysis, Speech, HealthKit, WatchConnectivity, simd, Python, Unity, C#, Git.

03 PROFESSIONAL EXPERIENCE

APPLE

June 2019 — Present

Senior Interaction Designer & Systems Prototyper / Model Behaviourist · Cupertino, CA

Direct multi-disciplinary prototype programs across wearable, audio, and spatial platforms. Manage technical roadmaps, resource allocation, and cross-functional handoffs between design, core software engineering, machine learning, and human factors research teams.

GENERATIVE AI MODEL BEHAVIOUR & EVALUATION

- **Automated Multimodal Benchmarking Pipeline:** Architected an automated CLI evaluation tool that processes multimodal audio and text inputs against frontier foundation models, transcribing and outputting timestamped, comparable benchmark reports with automated LLM-as-judge scoring.
- **Foundation Model Behaviour Framework:** Formulated an ethological methodology for studying generative AI behavior, analyzing response consistency, prompt sensitivity, and drift across model versions and adapter configurations.
- **Demographic & Prompt Perturbation Studies:** Investigated demographic representation and compositional limits in generative image models, establishing empirical performance boundaries for negative prompting and spatial conditioning across person presentations.

SENSOR-DRIVEN & AUDIO-FIRST INTERACTION PROTOTYPING

- **Wrist-Worn Systems Architecture (watchOS & iOS):** Architected high-performance watchOS and iOS companion prototypes utilizing SwiftUI, HealthKit, and background WatchConnectivity with robust offline queueing and local token state management.
- **Audio-Spatial Guidance User Study:** Conceived and executed a Wizard of Oz human factors study investigating audio-only spatial coaching. Identified critical trust-calibration failure modes and spatial ambiguities between novice and expert cohorts; authored branching interactive audio prototypes to resolve user disorientation.
- **Conversational Audio Ergonomics:** Engineered conversational audio prototypes exploring real-time acoustic scene classification, low-latency audio pipelines, and spatial audio ergonomics.
- **Contextual Task Coaching Framework:** Authored comprehensive interaction specifications for hands-free contextual task guidance, detailing multi-sensor activation patterns, failure recovery states, privacy guardrails, and computational effort models.

03 PROFESSIONAL EXPERIENCE (CONTINUED)

APPLE (CONTINUED)

Cupertino, CA

COMPUTER VISION & SPATIAL COMPUTING

- **On-Device Homography & Perspective Rectification (iOS 18+):** Engineered a high-precision computer vision tool mapping known reference images in live camera scenes to 3×3 simd homography transforms using solely public Vision and CoreImage APIs, eliminating external spatial tracking dependencies for planar registration. Authored accompanying masterclass curriculum and technical documentation.
- **Audio-First Spatial Wayfinding Architecture:** Directed the architecture and development of six generations of audio-first spatial wayfinding prototypes, integrating spatial audio, headphone motion tracking, acoustic classification, and local foundation models; validated across 127 automated test suites.

INCUBATION SYSTEMS & CROSS-FUNCTIONAL LEADERSHIP

- **Computer Vision Experience Evaluation Framework:** Authored standardized qualitative and quantitative evaluation rubrics for multi-modal computer vision and interactive camera experiences, aligning cross-functional teams around shared objective criteria.
- **Autonomous Multi-Agent Incubation Pipeline:** Designed an automated multi-agent prototype incubation workflow featuring formal review gates and structured decision logging to validate engineering and research integrity.
- **Collaborative Spatial Canvas:** Built a collaborative spatial web canvas supporting real-time multi-user interaction, 2D/3D spatial nodes, automated regression testing, and zero-trust local data isolation boundaries.
- **Sensor Feasibility & Latency Modeling:** Modeled interaction latency, computational constraints, and spatial tracking feasibility across conceptual sensor topologies to establish engineering evaluation rubrics.

KINIFIGAMES / BACKBEACH GAMES

2012 — May 2020

Co-Founder & Creative Director

- Co-founded independent studio; directed systems design and production for arcade titles including *Project4* (mechanics modeled on cognitive control and stimulus-response).
- Organized developer initiatives, coordinated at-cost convention showcases for indie creators, and represented women in games on national panels.

UNITY TECHNOLOGIES

September 2015 — May 2018

Developer Advocate · Software Test Engineer (QA) · Community Lead · San Francisco, CA

- Advanced from community management to software QA and global developer advocacy, representing pre-release features and triaging feedback to core runtime teams.
- Led cross-platform QA regression suites across mobile and desktop, maintaining strict confidentiality protocols for unannounced technologies and hardware partnerships.

04 GRANTED PATENTS & PUBLISHED APPLICATIONS

U.S. Patent 12,281,912: *User Interfaces for Maps and Navigation* — Application layouts for live routing updates and navigational overlays. **Granted 2025**

U.S. Patent 11,768,083: *User Interfaces for Providing Navigation Directions* — Dynamic canvas scaling and graphical overlays for directional prompts. **Granted 2023**

U.S. App 20250164266: *Grouping Maneuvers for Display in a Navigation Presentation* — Algorithmic logic grouping complex route maneuvers into stacked units. **Published 2025**

U.S. App 20240402211: *Location Measurement Techniques* — Position adjustments combining wireless ranging sessions with inertial tracking. **Published 2024**

European App EP4762325A1: *Systems and Methods for Presenting Information* — Cross-platform visualization of real-time telemetry and spatial positioning. **Published 2024**

05 RESEARCH PUBLICATIONS & SPEAKING

- **Interactive Research Explainers (Concept Field Guides):** Authored visual research distillations on *Trust in AI*, *Ethological AI Model Evaluation*, *Associative Memory*, and *Biomimetic Machine Learning*.
- **Conference Talks & Papers:** *Accessibility & Game Difficulty By Design* (2018); *Collaborate – Keeping the User Safe* (2017); *Random Generation and Feedback* (2014); Technical guides (*Collaborate User Guide*, *Bug Squashing*).

06 EDUCATION & FOUNDATIONS

INTERACTIVE SCIENCES, SYSTEMS MANAGEMENT & GAME DESIGN

Grounded in behavioural ecology, cognitive ergonomics, spatial orientation, and biomimetic human-computer interaction.

FOUNDATIONAL DISCIPLINES